University Research Management System

A university is developing a comprehensive database system to manage its research activities, faculty researchers, publications, and funding across multiple departments. The system needs to track who conducts research, what projects are being undertaken, how they are funded, and what academic outputs are produced.

- Each researcher in the system has a unique researcher ID, email address, and date of birth. The researcher's full name is stored as separate components: first name, middle name, and last name.
- Each researcher has an office location, which consists of a building name and room number.
- Since researchers often need to be contacted through multiple channels, the system stores multiple phone numbers for each researcher (such as office phone, mobile phone, and lab extension).
- Researchers specialize in various research areas, and a single researcher may have expertise in multiple fields such as "Artificial Intelligence","Machine Learning", "Database Systems", or "Computer Vision".
- The system should automatically calculate each researcher's current age based on their date of birth.

Within the research community, senior researchers often supervise junior researchers and doctoral students:

- This supervisory relationship must be tracked carefully, including when the supervision began and the nature of the supervision (whether it's a mentor, advisor, or co-advisor relationship).
- A researcher can supervise multiple junior researchers, but each junior researcher typically has only one primary supervisor.
- Some senior researchers may not have supervisors themselves, while junior researchers usually do.

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The university manages numerous research projects:

- Each identified by a unique project ID. For each project, the system stores the project title, start date, expected end date, total budget allocation, and current status (such as "Active", "Completed", or "Pending").

- Projects are often described by multiple research keywords that help categorize and search for them, such as "Deep Learning", "Natural Language Processing", or "Medical Diagnosis".

- Many projects receive funding from multiple sources, so the system must track all funding agencies associated with each project, including organizations like "National Science Foundation", "National Institutes of Health", or private foundations.

- The system should automatically calculate the project duration in months based on the start and end dates.

Every research project must have exactly one lead researcher who serves as the principal investigator and takes primary responsibility for the project.

- However, a lead researcher may oversee multiple projects simultaneously. Beyond the lead investigator, multiple researchers can work on the same project in various capacities.

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- The system needs to track when each researcher joined a project, their specific role (such as "Lead Investigator", "Co-Investigator", or "Research Assistant"), how many hours per week they dedicate to the project, and what percentage of the project credit they receive.

- These details vary for each researcher on each project and must be recorded separately for every researcher-project combination.

Research outputs are published as publications :

- Teach with a unique publication ID. The system stores the publication title, publication date, DOI (Digital Object Identifier), publication type (Journal or Conference), and the current citation count. Each publication is associated with multiple keywords for indexing purposes.

- Since publications often have authors from multiple institutions, the system stores a complete list of all author names, which may include researchers from this university as well as external collaborators.

When researchers from this university author publications :

- the system must track each researcher's specific contribution.
- This includes their position in the author list (first author, second author, etc.) and the type of contribution they made (such as "Writing", "Methodology", "Data Analysis", or "Review"). A researcher can author multiple publications, and each publication typically has multiple authors.

Research funding comes from grants :

- each identified by a unique grant ID. The system stores the official grant name, total amount awarded, grant start date, grant end date, and the name of the funding agency providing the grant.
- The system should automatically calculate the grant period in months from the start and end dates.

Grants can fund multiple research projects, and conversely, a single project may receive funding from multiple grants :

- When a grant funds a project, the system must record how much money from that grant was allocated to the specific project and when this allocation was made.
- These allocation details are specific to each grant-project combination and cannot be stored with either the grant or project alone.

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